STA 312: Distribution Theory I

Lecture Notes: CLT & Bivariate MGFs

DR. C. C. NNANATU & DR. O. J. OBULEZI

Department of Statistics, UNIZIK

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1. CLT: Setup and Standardization

Let $X_1, X_2, \ldots, X_n$ be i.i.d. random variables with $\mathbb{E}[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$. We want to find the limiting distribution of:

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We can rewrite Z_n by expanding the sample mean $\bar{X}_n$:

$$Z_n = \frac{\frac{1}{n}\sum X_i - \mu}{\sigma/\sqrt{n}} = \sum_{i=1}^{n} \frac{1}{\sqrt{n}} \left(\frac{X_i - \mu}{\sigma} \right)$$

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Let $Y_i = \frac{X_i - \mu}{\sigma}$. The variables Y_i are still i.i.d., with:

$$\mathbb{E}[Y_i] = 0 \quad \text{and} \quad \text{Var}(Y_i) = \mathbb{E}[Y_i^2] = 1$$

Thus, our target simplifies to: $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} Y_i$

2. CLT: Taylor Expansion of the MGF

The MGF of a single standardized variable Y_i is $M_Y(t) = \mathbb{E}[e^{tY}]$. Expanding $M_Y(t)$ as a Taylor series around $t = 0$:

$$M_Y(t) = \mathbb{E}[1] + \mathbb{E}[Y]t + \frac{\mathbb{E}[Y^2]}{2!}t^2 + \frac{\mathbb{E}[Y^3]}{3!}t^3 + \dots$$

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Substituting our known moments $\mathbb{E}[Y] = 0$ and $\mathbb{E}[Y^2] = 1$:

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Now, because the Y_i variables are independent, the MGF of their sum Z_n is the product of their individual MGFs:

$$M_{Z_n}(t) = \mathbb{E}\left[e^{\frac{t}{\sqrt{n}}\sum Y_i}\right] = \left[M_Y\left(\frac{t}{\sqrt{n}}\right)\right]^n$$

3. CLT: Taking the Limit

Substituting our Taylor expansion into the MGF of Z_n:

$$M_{Z_n}(t) = \left[1 + \frac{\left(\frac{t}{\sqrt{n}}\right)^2}{2} + \mathcal{O}\left(\left(\frac{t}{\sqrt{n}}\right)^3\right)\right]^n = \left[1 + \frac{t^2}{2n} + \mathcal{O}\left(\frac{t^3}{n^{3/2}}\right)\right]^n$$

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Proof Completion.

Taking the limit as $n \to \infty$, and recalling that $\lim_{n\to\infty}(1 + \frac{c}{n})^n = e^c$:

$$\lim_{n\to\infty} M_{Z_n}(t) = e^{\frac{t^2}{2}}$$

Since $e^{\frac{t^2}{2}}$ is the MGF of a standard normal distribution $\mathcal{N}(0,1)$, by the Continuity Theorem, $Z_n \xrightarrow{d} \mathcal{N}(0,1)$. □

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Proof.

Let $f(x, y)$ be the joint PDF of continuous random variables X and Y.

$$\begin{aligned} M_{X,Y}(t_1, 0) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{t_1 x + (0) y} f(x, y)\, dx\, dy \\ &= \int_{-\infty}^{\infty} e^{t_1 x} \left(\int_{-\infty}^{\infty} f(x, y)\, dy \right) dx \\ &= \int_{-\infty}^{\infty} e^{t_1 x} f_X(x)\, dx = \mathbb{E}[e^{t_1 X}] = M_X(t_1) \end{aligned}$$

Where $f_X(x)$ is the marginal density of X. □

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$$M_{X,Y}(t_1, t_2) = \mathbb{E}\left[e^{t_1 X + t_2 Y}\right] = \mathbb{E}\left[e^{t_1 X} \cdot e^{t_2 Y}\right]$$

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Since X and Y are independent, any functions $g(X)$ and $h(Y)$ are also independent. Thus, $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$:

$$M_{X,Y}(t_1, t_2) = \mathbb{E}\left[e^{t_1 X}\right] \cdot \mathbb{E}\left[e^{t_2 Y}\right] = M_X(t_1) \cdot M_Y(t_2)$$



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Differentiating the joint MGF with respect to t_2:

$$\frac{\partial M_{X,Y}}{\partial t_2} = \frac{\partial}{\partial t_2}\mathbb{E}\left[e^{t_1 X + t_2 Y}\right] = \mathbb{E}\left[Y e^{t_1 X + t_2 Y}\right]$$

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Now, differentiating this result with respect to t_1:

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Evaluating the cross-partial derivative at $t_1 = 0, t_2 = 0$:

$$\left.\frac{\partial^2 M_{X,Y}}{\partial t_1 \partial t_2}\right|_{(0,0)} = \mathbb{E}\left[XY e^0\right] = \mathbb{E}[XY]$$